



GM University

Davanagere, Karnataka, India

DEPARTMENT OF CYBER SECURITY & INFORMATION SECURITY

Course Code : UE24CS2306 -PROJECT BASED LEARNING

Project Title : Health Monitoring System Using IOT

Semester & Year : 3rd Sem & 2nd year

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**Name of Supervisor/
Name of co-Supervisor**

Date:

**Supervisor Name and Signature
(Nayana M R)**

		GM University Davanagere, Karnataka, India Department of Cyber security and Information Security	
Program:	B-TECH		
Department:	CY & IY	Session:	2025-26

Project Title:	Health Monitoring System using IOT
Project Abstract: The Health Monitoring System is an IoT-based project that continuously tracks vital health parameters such as heart rate, SpO₂, temperature, and humidity using an ESP32 microcontroller with MAX30102 and DHT22 sensors. The collected data is transmitted via Wi-Fi to a Node.js backend and visualized in real time on a web dashboard using Chart.js, with an option to export data to CSV or Excel for analysis. Designed as a low-cost, open-source solution, this system enables remote and contactless health monitoring, providing early alerts for abnormal readings and supporting better healthcare management.	
Goals & Objectives: ▪ Continuously measure heart rate, SpO₂, temperature & humidity with ESP32. ▪ Transmit data over Wi-Fi (MQTT / HTTP) to lightweight Node.js backend. ▪ Visualize live metrics on responsive web dashboard (Chart.js). ▪ One-click export of historical data to CSV / Excel.	
Project Scope: 1. Integration with Cloud Platforms: Extend the system to cloud services such as AWS IoT or Google Cloud to enable remote monitoring, data analytics, and long-term record storage. 2. Mobile Application Development: Develop an Android/iOS app for real-time alerts, personalized health insights, and access to data anytime, anywhere. 3. AI-Based Health Prediction: Implement machine learning algorithms to predict potential health risks (like cardiac events or fevers) from sensor trends. 4. Multi-User and Doctor Dashboard: Add role-based dashboards allowing doctors to monitor multiple patients simultaneously and receive automatic alerts for abnormal readings. 5. Wearable Integration: Miniaturize the system into a wearable device or smartwatch for continuous, portable health monitoring. 6. Enhanced Security & Privacy: Incorporate encryption and authentication mechanisms to ensure safe transmission and storage of sensitive health data.	

Tools / Technologies to be used in Project:

- ESP32-WROOM-32 – dual-core 240 MHz, Wi-Fi/BT, 18 ADC, 3.3 V.
- DHT22 (upgrade to DHT11) – ± 0.5 °C, 0-100 % RH, 1-wire digital.
- MAX30102 – I²C pulse-oximeter & HR, 600 nm RED + 880 nm IR.
- Breadboard + jumper wires – rapid prototyping.
- USB-C cable – power + programming.
- Optional – 0.96" OLED (I²C), 3.7 V Li-ion pack, DS3231 RTC.

Supervisor Information

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